

Lab 3: Component Modelling & Architectural Pattern Selection

Restaurant Table Booking & Pre-Ordering App

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1. Architecture Selection

We chose Layered Architecture for the Restaurant Table Booking & Pre-Ordering App.

2. Architectural Choice

The application has clearly separated responsibilities for user interaction, table and reservation management, order processing, menu and preparation handling, and payment processing. A layered structure organizes these responsibilities into clear presentation, business, and data-related responsibilities.

3. Scenario-Specific Reasons

- **Clear separation of concerns:** The restaurant app manages table availability and reservations, pre-orders, preparation timelines, and payments. Separating these responsibilities makes the system easier to understand, maintain, and modify.
- **Suitable application structure:** The core operations are closely related within one restaurant system, so a layered design provides a structured solution without the additional operational and communication complexity of independently deployed microservices.

4. Security Advantage

The presentation layer can be separated from payment and reservation-management functionality. Access controls can therefore be applied between layers so that only authorized operations can modify payment, table, and reservation information. This supports the Lab 1 requirement that only authenticated restaurant managers can modify table and reservation statuses.

5. Performance Benefit

The layered design allows related business operations such as table-status validation, reservation updates, order processing, and preparation-time calculation to communicate through defined internal interfaces. This avoids unnecessary distributed communication for closely related operations and supports responsive processing of table-status updates.